Embeddings and language models

AI training workshop

Rahul Dave

How a language model generates text

It turns pieces of text into numbers.

It uses the text around each piece.

It predicts what text can follow.

A brief history

2013 Word embeddings capture relationships.

2017 Transformers use attention to process context.

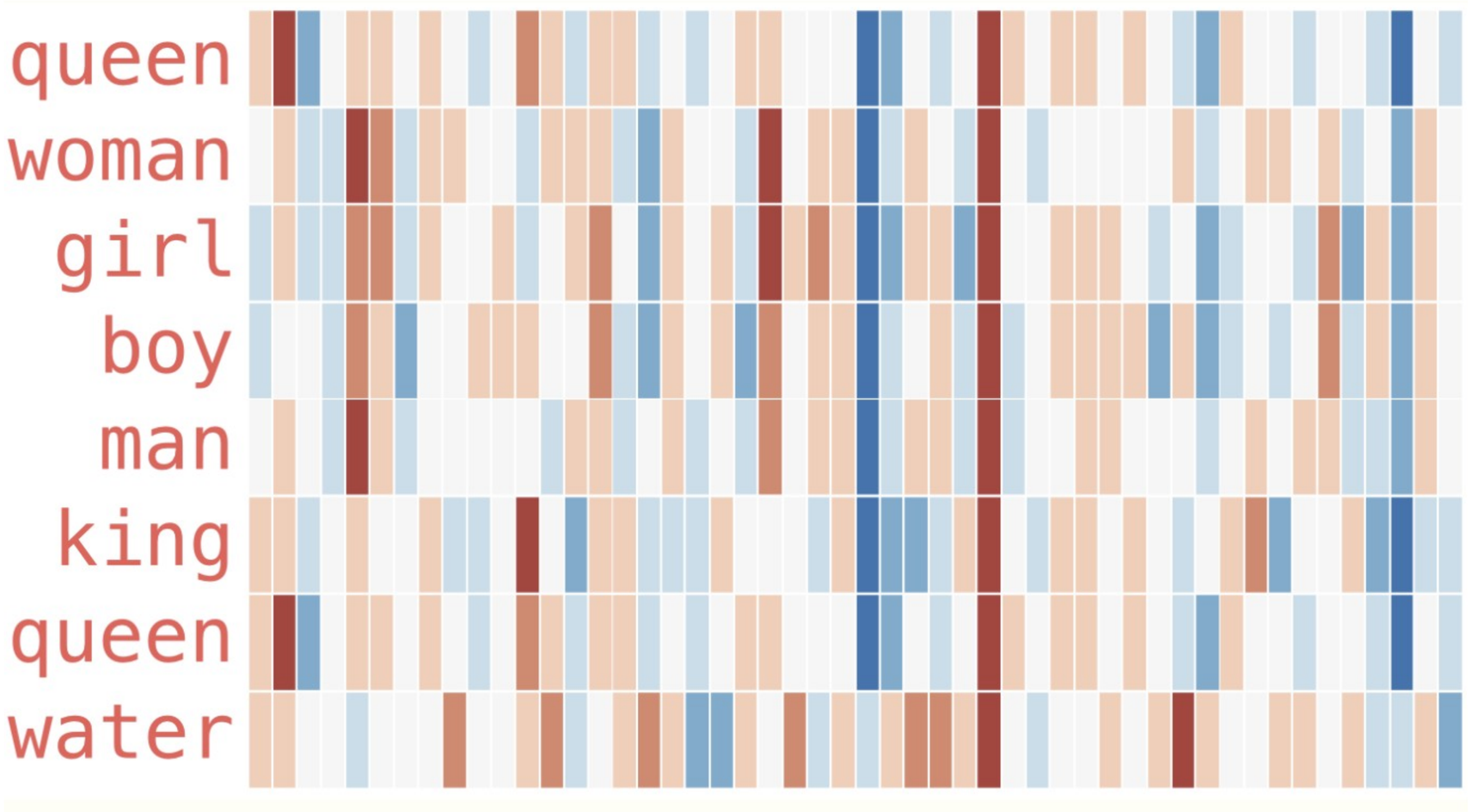
The dates mark selected milestones in a much longer history.

An embedding is a list of numbers

Models learn the numbers from examples.

The numbers help compare words and passages.

Words as numbers



Visual from Rahul Dave's original teaching deck

Words with related meanings



Nearby points illustrate similar word use.

Visual from Rahul Dave's original teaching deck

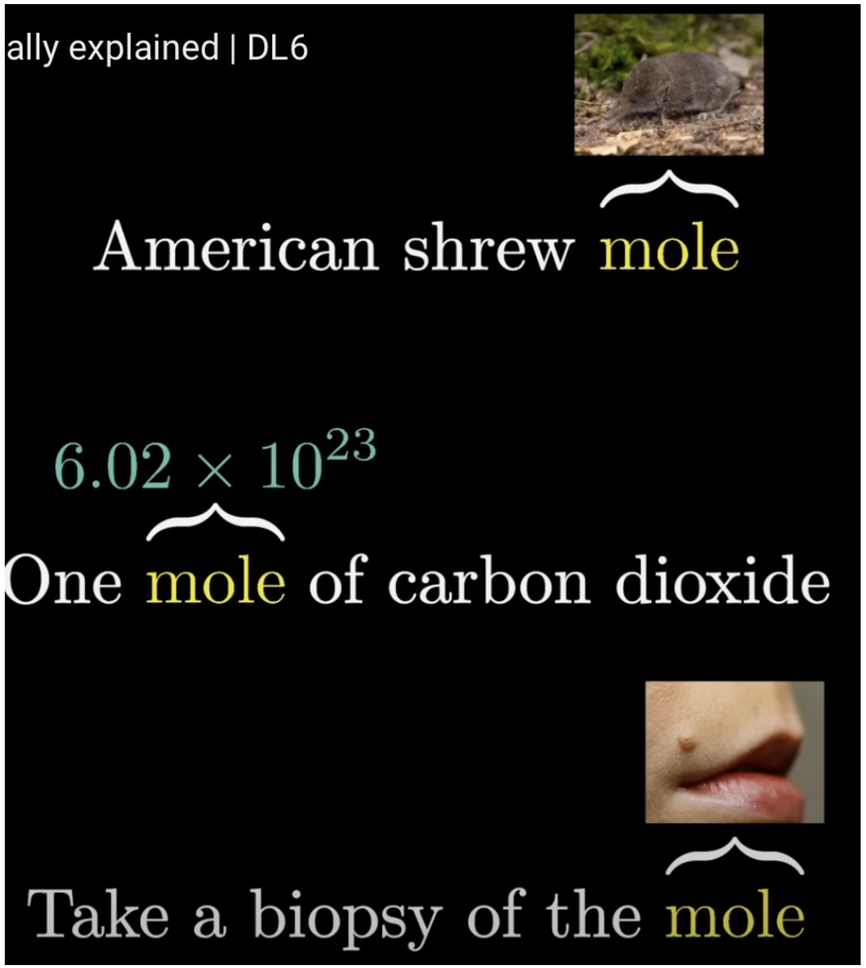
The word "bank" depends on context

The river bank was muddy.

The bank approved the loan.

Context is the text the model has for its current response.

One word, several meanings



Context changes

the meaning of "mole".

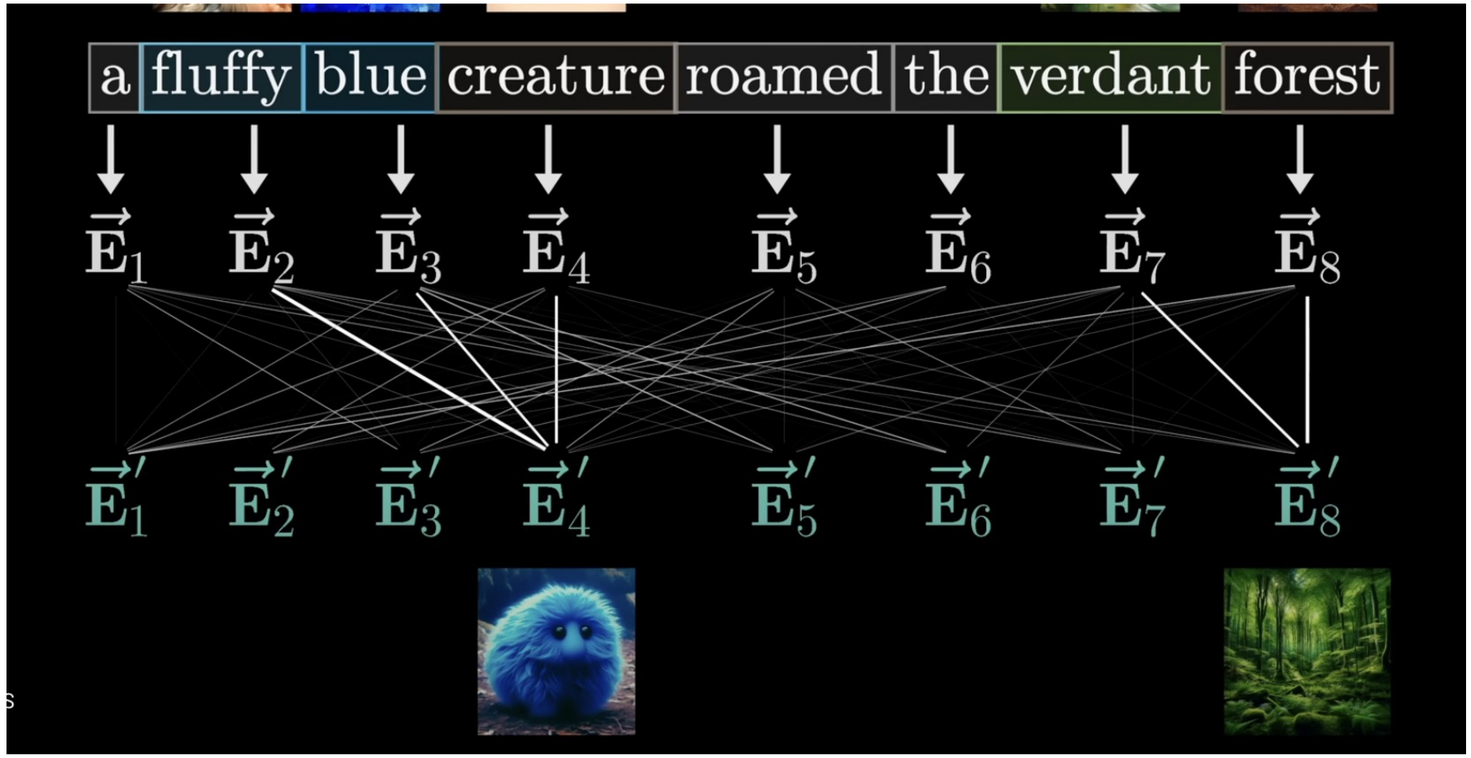
Visual: Grant Sanderson, 3Blue1Brown

Attention uses the surrounding text

Attention combines information across the text.

The result depends on the current context.

The words describe different things



Fluffy and blue

describe the creature.

Verdant means green.

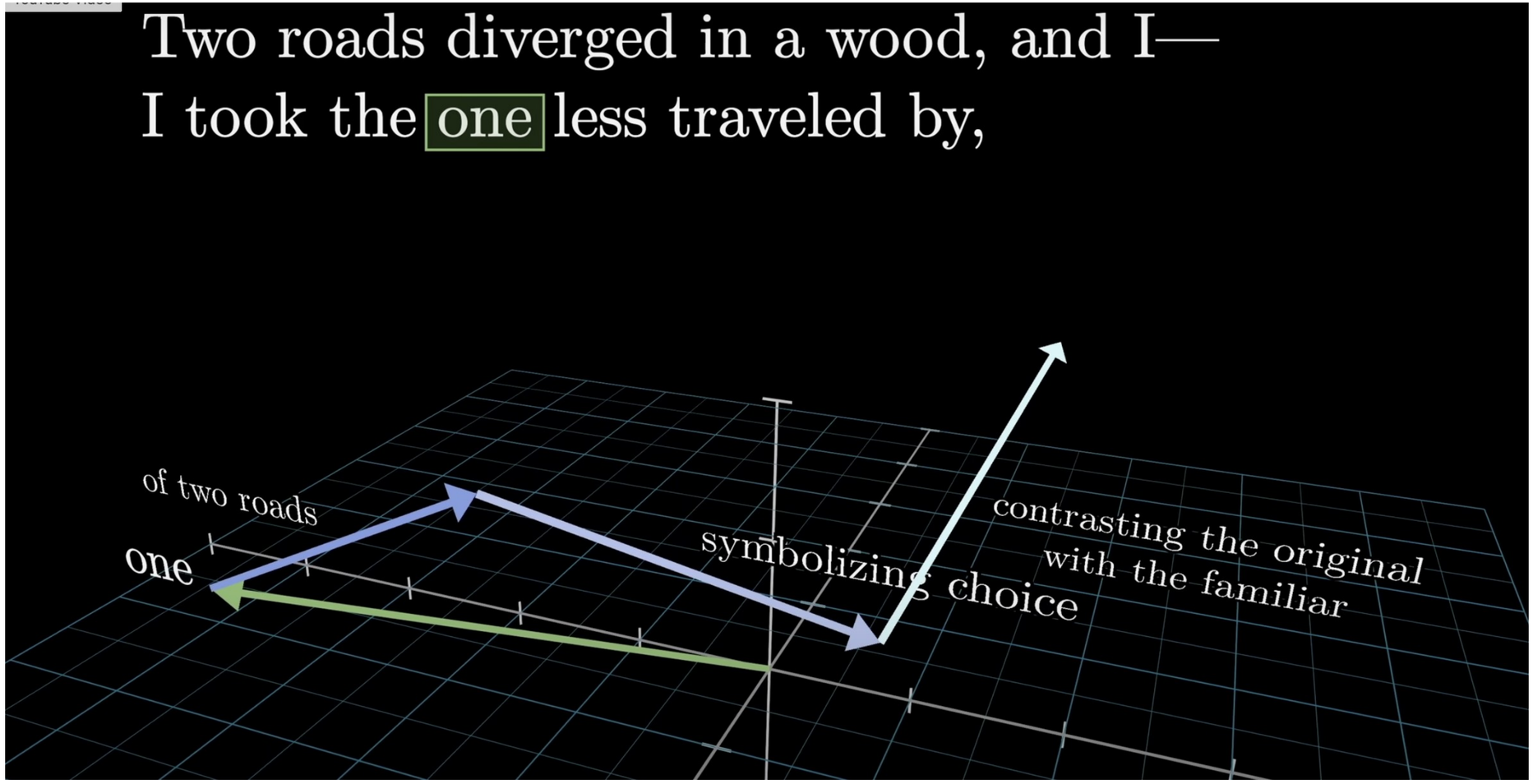
Visual: Grant Sanderson, 3Blue1Brown

Training also stores learned associations

Michael Jordan plays the sport of ...

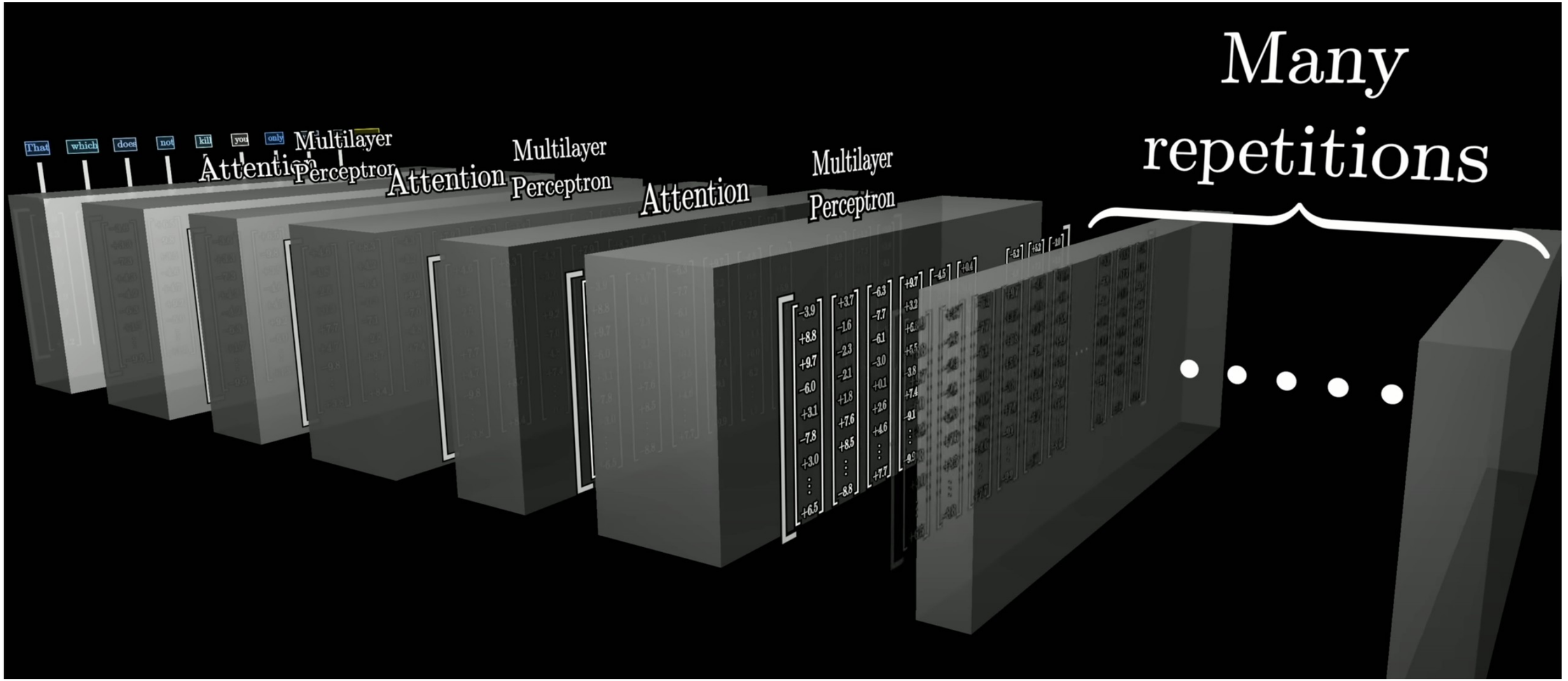
The model can associate the name with basketball.

Context can add several kinds of meaning



Visual: Grant Sanderson, 3Blue1Brown. Poem: Robert Frost.

The model repeats these operations



Visual: Grant Sanderson, 3Blue1Brown

Learning from text

During training, the model predicts what text comes next.

Prediction errors guide changes in the model.

Training changes the model. A prompt supplies context for a response.

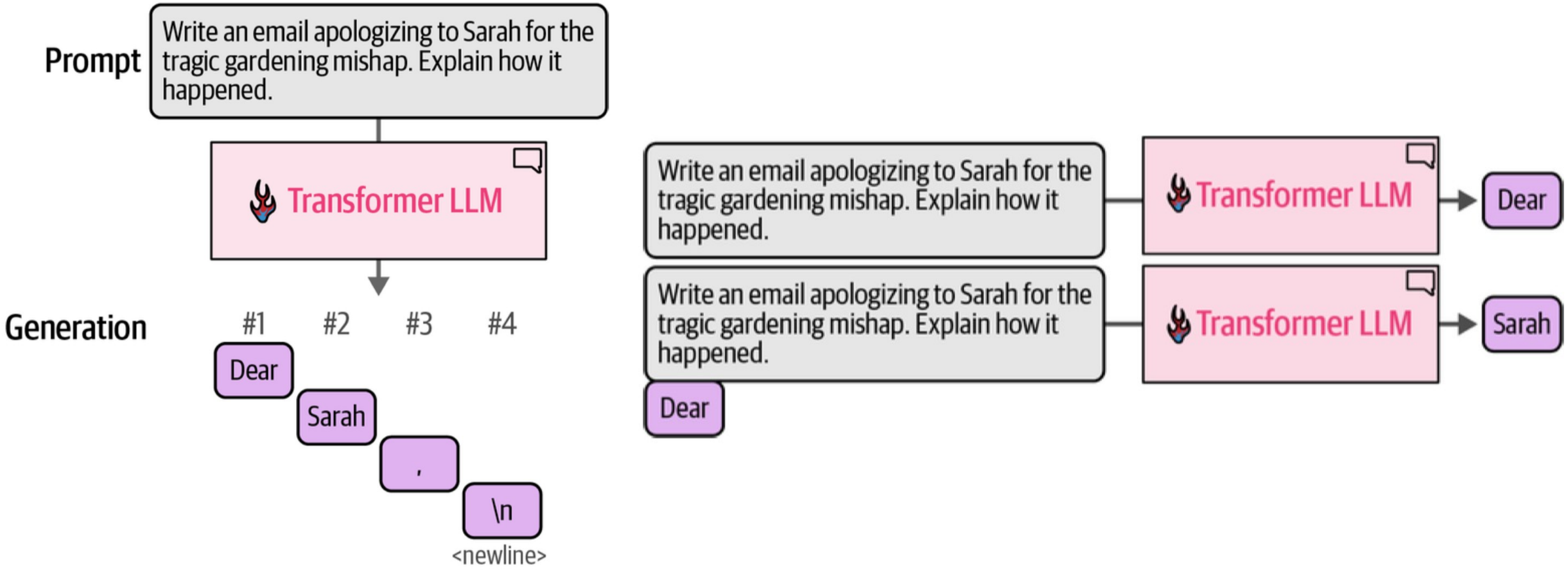
A token is a piece of text

A token can be a word or part of a word.

Punctuation can also be a token.

A language model generates a response one token at a time.

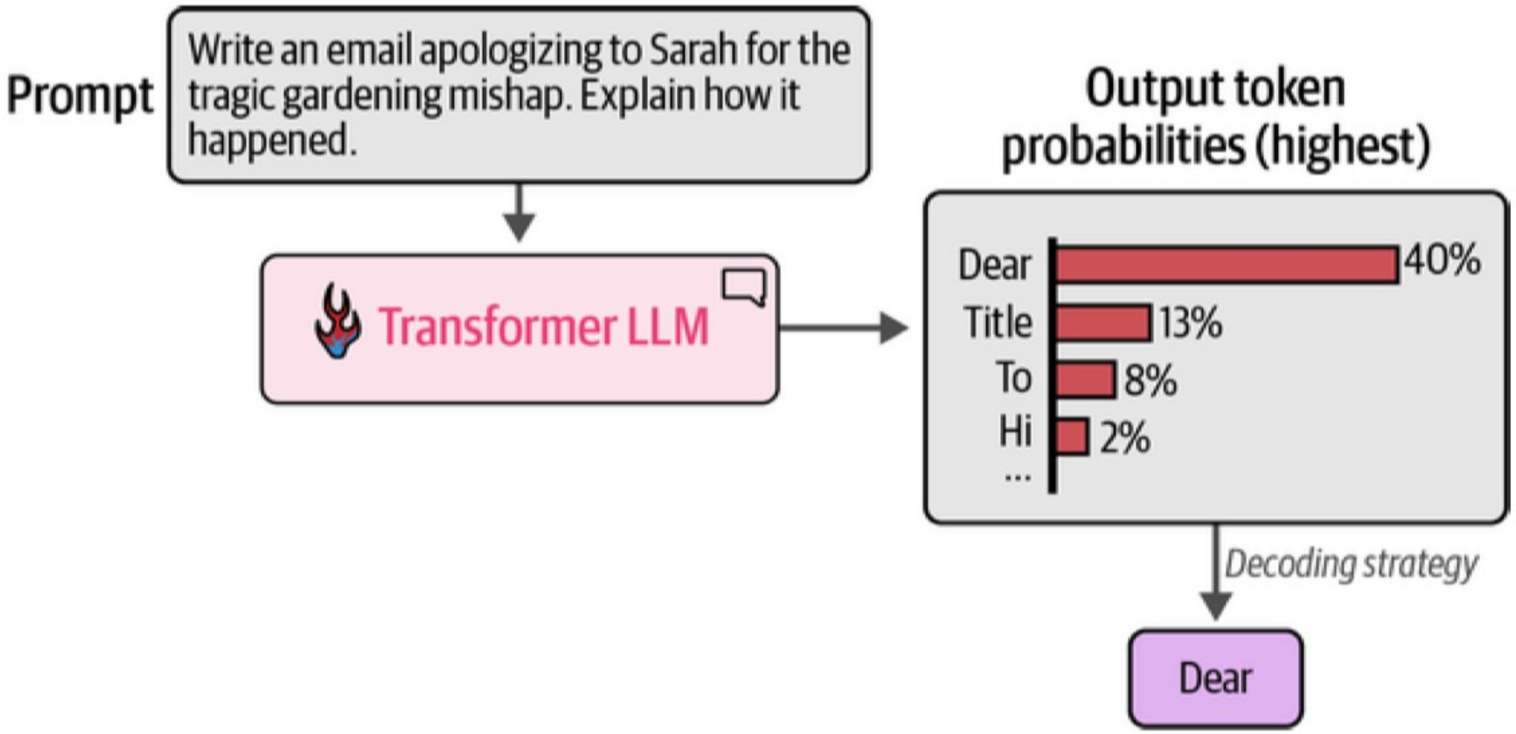
Generation happens token by token



Visual: Alammar et al., from Rahul Dave's original deck

Probabilities drive everything

We sample from probabilities to get the next word



Visual: Alammar et al., from Rahul Dave's original deck

Sentence completion

The train arrived at the ...

Suggest a next word. Then explain which context would change your answer.

The text we send shapes the response

A prompt is the text we send to the model.

Relevant details can change the answer.

A fluent answer can still be wrong

Text prediction does not verify a claim.

Important answers need checks against reliable sources.

The choices we make with AI

We choose the task and the context.

We decide how to check the answer.

Next session: the AI ecosystem we want for ourselves.

Sources and further viewing

Rahul Dave
Original teaching decks and Credcore notes

3Blue1Brown: Transformers

3Blue1Brown: Attention

3Blue1Brown: Learned facts